ZCAS University

CIT3641 – ADVANCED DATABASES

MID-SEMESTER TEST

Wednesday, 22nd April 2026

10:30 Hrs – 12:00 Hrs

TIME ALLOWED: WRITING – 1.5 HOURS

READING TIME – 5 MINUTES

INSTRUCTIONS:

1. All the twenty multiple-choice questions in **Section One** (Question One) are **compulsory** and must be attempted. ONLY a letter A, B, C, or D should be indicated as an answer for each multiple-choice question. NO explanations are required. Indicating 2 or more letters on a question will result in a zero mark for that question.
2. **Section Two** has two questions (Question Two and Question Three), **ONLY ONE** of the two questions should be attempted.
3. Maximum score for each Section is 20 marks: Giving a total of 40 marks for the whole test.
4. Numbering of answers in your answer booklet should be the same as that in the question paper. Answers that are not numbered will NOT be marked.
5. Use of scientific calculators during the test is NOT allowed.
6. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION ONE – Answer ALL the 20 Multiple Choice Questions
(Each question carries one (1) mark)

1. What does SQL stand for?
   A. Sequential Query Language
   B. Structured Query Language
   C. Standard Query Language
   D. Software Query Language

2. What is a database management system (DBMS)?
   A. A collection of related tables
   B. Software for creating and managing databases
   C. A language for querying databases
   D. A hardware component for storing data

3. Which of the following is NOT a classification of database systems?
   A. Relational
   B. Hierarchical
   C. NoSQL
   D. Sequential

4. What is metadata in the context of databases?
   A. Data stored in tables
   B. A type of encryption key
   C. Information about data
   D. Data modeling technique

5. What is data modeling?
   A. Creating diagrams to represent database structure
   B. Storing data in tables
   C. Querying databases using SQL
   D. Encrypting sensitive information

6. What is entity-relationship modeling used for?
   A. Optimizing database performance
   B. Defining entities and relationships among them
   C. Generating SQL queries
   D. Creating data dictionaries

7. What is relational modeling?
   A. Storing data in a hierarchical structure
   B. Encoding data for secure transmission
   C. Using flat files to organize data
   D. Representing data using tables and relationships

8. What is the purpose of data normalization?
   A. Deleting redundant data
   B. Inserting new records into a database
   C. Updating existing data
   D. Organizing data to minimize redundancy and dependency

9. What is the primary key in a relation?
   A. A key attribute used for indexing
   B. The first attribute in the relation
   C. A set of attributes that uniquely identifies tuples
   D. The attribute with the highest cardinality

10. What is a foreign key in a relation?
   A. A key attribute used for indexing
   B. An attribute that references the primary key of another relation
   C. A set of attributes that uniquely identifies tuples
   D. The attribute with the highest cardinality

11. What is the purpose of the COMMIT statement in SQL?
   A. To save changes made to the database
   B. To roll back changes made to the database
   C. To delete records from a table
   D. To create a new table in the database

12. What is the purpose of the ROLLBACK statement in SQL?
   A. To save changes made to the database
   B. To roll back changes made to the database
   C. To delete records from a table
   D. To create a new table in the database

13. In an Entity-Relationship Diagram (ERD), what does a mandatory relationship indicate?
   A. The relationship is optional
   B. The relationship is required
   C. The relationship involves multiple entities
   D. The relationship is bidirectional

14. In an ERD, what does an optional relationship indicate?
   A. The relationship is mandatory
   B. The relationship is required
   C. The relationship is not always necessary
   D. The relationship is bidirectional

15. What is a fan trap in database modeling?
   A. An error in the SQL syntax
   B. A situation where multiple relationships converge on a single entity
   C. A problem that arises from violating normalization rules
   D. A mistake in the data dictionary

16. What is the difference between a determinant and a candidate key?
   A. They are synonymous terms
   B. A determinant uniquely determines other attributes in a relation, while a candidate key is a minimal set of attributes that uniquely identifies tuples
   C. A determinant is a key attribute, while a candidate key is a non-key attribute
   D. A determinant is used for sorting data, while a candidate key is used for indexing

17. When converting a mandatory one-to-one relationship into relations, how many tables are typically created?
    A. One
    B. Two
    C. Three
    D. None

18. When converting a one-to-many relationship into relations, which side of the relationship becomes the foreign key?
    A. The "one" side
    B. The "many" side
    C. Both sides
    D. Neither side

19. When converting a many-to-many relationship into relations, how many tables are typically created?
    A. One
    B. Two
    C. Three
    D. None

20. What is the purpose of referential integrity constraints in a database?
    A. To ensure that each tuple in a relation is unique
    B. To enforce the database index rule
    C. To maintain consistency between related tables
    D. To optimize query performance

SECTION TWO – Answer ONE of the 2 Questions below
(Each question carries twenty (20) marks)

QUESTION TWO

Question 2 – Part One **(10 Marks)**
Workers in the town stay in Lodgings. A lodging may have many workers in it. We are only concerned with the workers' current lodging. Workers possess certain Skills such as 'Tree Surgery', and 'Garden Design'. They may have more than one skill.

Based on the above information, you are required to draw an ER Diagram – showing identified entities, relationships between them and a primary key for each entity.

Question 2 – Part Two **(10 Marks)**
Using appropriate database tables, explain how a multi-value attribute, such as mobile phone numbers, may be represented in a database / table?

You are required to draw database tables which will provide for a situation where data about students without any mobile number, students with only one mobile number and those with more than one mobile number may be provided for in the database.

QUESTION THREE

1. Explain the most likely output from the SQL Command below: **(5 Marks)**

```
CREATE TABLE EMPTEMP
   AS SELECT * FROM EMP
   WHERE 1 = 2;
```

2. Explain the most likely output from the SQL Command below: **(5 Marks)**

```
DELETE * FROM EMPLOYEE;
```

3. State the most likely function / role for each of the five (5) keywords (FROM, WHERE, GROUP, HAVING, ORDER) in the SQL Command below:

(5 Marks)

```
SELECT branchID, COUNT(staff_id)
FROM worker
WHERE branchType = 'Main'
GROUP BY branchID
HAVING COUNT (staff_id) > 1
ORDER BY branchID;
```

NOTE:

NO essay type of answering. Pick each keyword and separately state its function / role in one or two sentences.

4. Using the "worker" table below and the SQL Command in question 3 above, determine the final output that will result from running / executing the SQL Command.

NOTE:

Show the step-by-step execution of the SQL Command, leading to the final output.

(5 Marks)

worker table

staff_id	staffName	branchID	branchType
101	Alice	B01	Main
102	Bob	B01	Main
103	Carol	B02	Main
104	David	B02	Sub
105	Emma	B03	Main
106	Frank	B03	Main
107	Grace	B03	Main
108	Henry	B04	Sub
109	Irene	B04	Main

END OF TEST